Trading the ERCOT North power premium

Henry Hub gas economics, day-ahead pricing and real-time settlement risk

TRADE DECISION / NOVEMBER 6 DELIVERY

Stay flat. The model favors selling day-ahead and buying real-time by $0.93/MWh. The assumed $1.00/MWh round-trip cost consumes that edge.

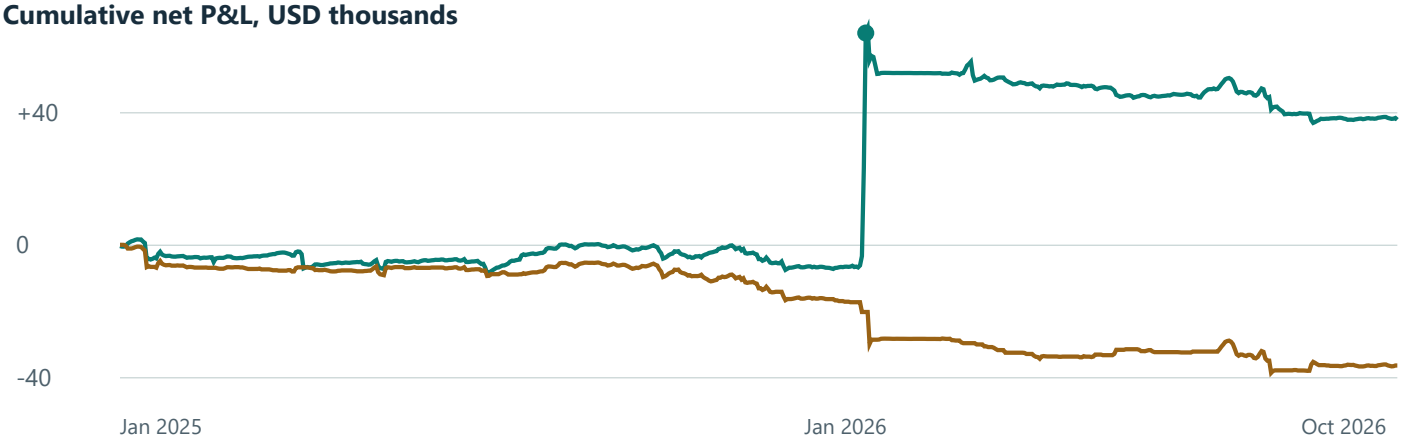
MARKET THESIS

A gas plant with a heat rate of 7 MMBtu/MWh needs another $7/MWh of power revenue when gas rises $1/MMBtu. I tested whether the resulting spark spread, recent ERCOT premiums and DFW weather could identify when day-ahead power was expensive relative to real-time power.

HISTORICAL PAPER P&L

669 delivery days / Jan 2025 to Oct 2026

Cumulative net P&L, USD thousands



Always short day-ahead +$37,912 **Weather model -$36,295**

WHAT DROVE THE RETURN

On January 26, day-ahead power averaged $694.03/MWh against $167.42/MWh in real time. Selling day-ahead and buying back 77.62 MWh in real time earned $40,799 after costs. Excluding that day, the short benchmark lost $2,887. The weather model held zero exposure that day.

DID WEATHER IMPROVE THE TRADE?

On January 28, the model using weather sold 77.6 MWh day-ahead at $76.65/MWh and bought it back in real time at $199.89/MWh. The trade lost $9,641 after costs. The model without weather stayed flat. Across all 669 days, adding weather inputs lowered net trading results by $26,250.

RESEARCH IMPLEMENTATION

Python refreshes public feeds and fits each forecast on 180 prior observations. SQLite stores settlements and dated paper signals. Excel reconciles positions, fees and daily P&L. Fills are hypothetical.

Project code and results / github.com/SourRanger/ercot-north-trading

ERCOT settlement archives EIA Henry Hub via FRED Iowa Environmental Mesonet